

01:960:490 - Homework 2

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1. 3.11: The tensile strength of Portland cement is being studied. Four different mixing techniques can be used economically. A completely randomized experiment was conducted and the following data were collected:

Mixing technique	Tensile strength (lb/in ²)			
1	3129	3000	2865	2890
2	3200	3300	2975	3150
3	2800	2900	2985	3050
4	2600	2700	2600	2765

- (a) Test the hypothesis that mixing techniques affect the strength of the cement. Use $\alpha = 0.05$.
- (b) Construct a graphical display as described in Section 3.5.3 to compare the mean tensile strengths for the four mixing techniques. What are your conclusions?
- (c) Use the Fisher least significant difference (LSD) method with $\alpha = 0.05$ to make comparisons between pairs of means.
- (d) Construct a normal probability plot of the residuals. What conclusion would you draw about the validity of the normality assumption?
- (e) Plot the residuals versus the predicted tensile strength. Comment on the plot.
- (f) Prepare a scatter plot of the results to aid the interpretation of the results of this experiment.

Solution: We use a one-way ANOVA with factor “mixing technique”:

$$Y_{ij} = \mu + \tau_i + \varepsilon_{ij}, \quad i = 1, \dots, 4, \quad j = 1, \dots, 4,$$

with independent $N(0, \sigma^2)$ errors.

(a) Hypotheses:

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4 \quad \text{vs} \quad H_a : \text{not all } \mu_i \text{ are equal.}$$

From `aov(strength ~ mixing)` we get

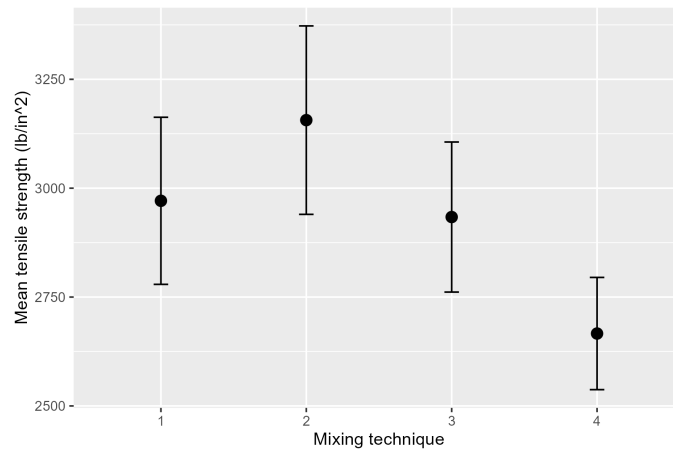
Source	Df	SS	MS	<i>F</i>
Mixing	3	489,740	163,247	12.73
Error	12	153,908	12,826	

with $p = 0.000489 < 0.05$. So we reject H_0 : mixing technique affects mean tensile strength.

(b) The sample means (lb/in²) are

$$\bar{y}_1 = 2971, \quad \bar{y}_2 = 3156.25, \quad \bar{y}_3 = 2933.75, \quad \bar{y}_4 = 2666.25.$$

The mean–CI plot below shows techniques 1 and 2 clearly above 4, with 3 in between. Techniques 1 and 2 are the high-strength group, 3 is moderate, and 4 is lowest.

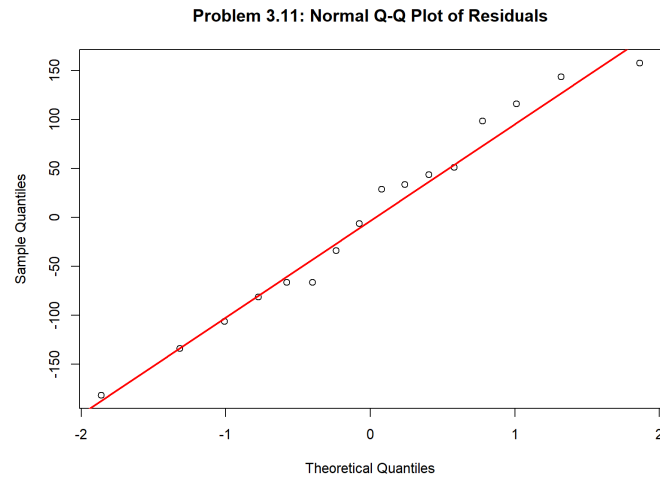


(c) Using $MSE = 12,826$ and $df_E = 12$, Fisher's LSD via `pairwise.t.test` gives

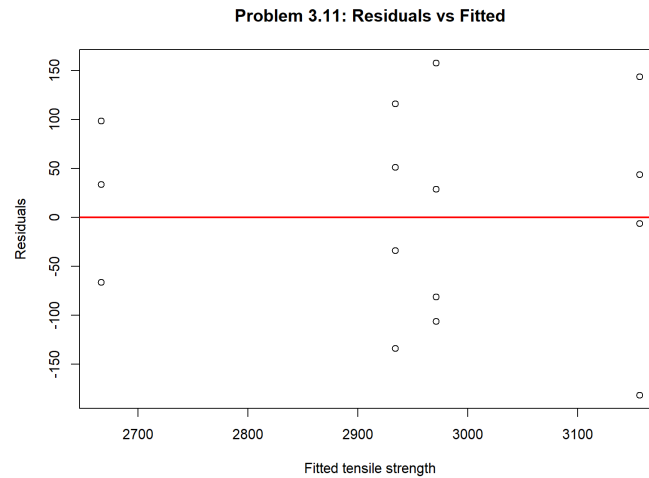
	1	2	3
2	0.0392		
3	0.6501	0.0167	
4	0.0025	0.000052	0.0059

At $\alpha = 0.05$, 1 and 2 differ (0.0392), 1 and 3 do not (0.6501). Technique 4 differs from 1, 2, and 3; 2 also differs from 3. So 2 tends to be highest, 4 clearly lowest, with 1 and 3 in the middle.

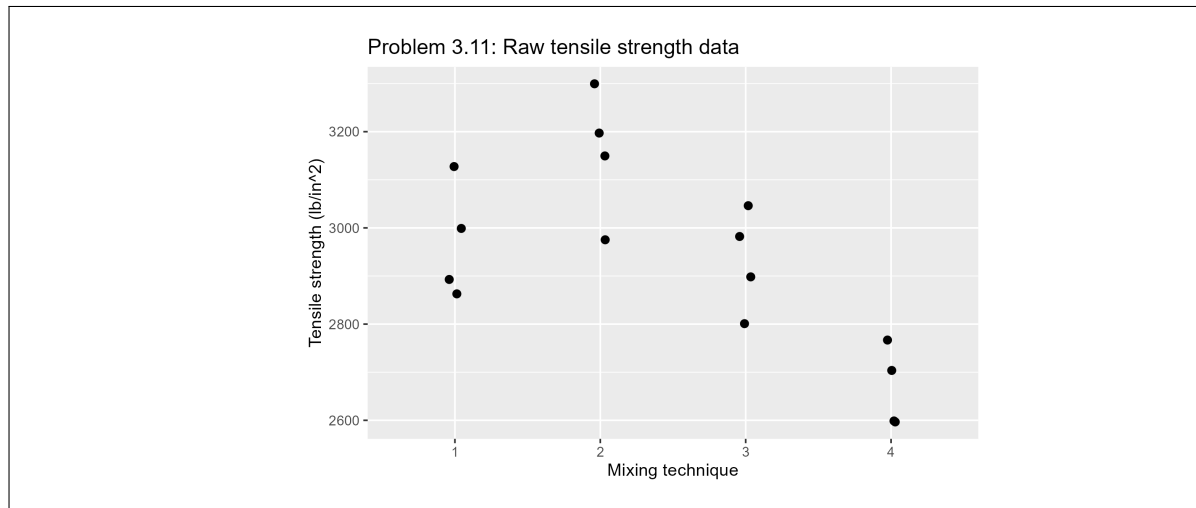
- (d) The normal Q–Q plot is roughly linear with only slight tail deviation, so the normality assumption looks fine.



- (e) Residuals vs fitted values are spread roughly evenly around zero with no pattern or change in spread, so constant-variance and model form look reasonable.



- (f) The raw-data plot shows four clusters by technique: 2 has the largest values, 4 the smallest, and 1–3 in between. There are no obvious outliers, and the plot agrees with the ANOVA and pairwise results.



2. 3.12:

- Rework part (c) of Problem 3.11 using Tukey's test with $\alpha = 0.05$. Do you get the same conclusions from Tukey's test that you did from the graphical procedure and/or the Fisher LSD method?
- Explain the difference between the Tukey and Fisher procedures.

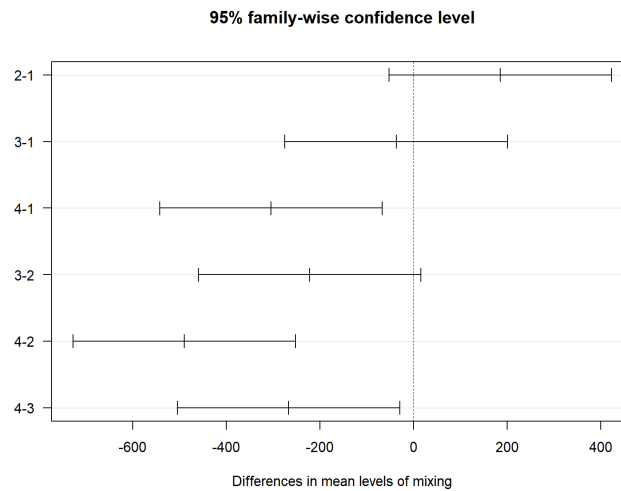
Solution:

(a) Applying Tukey's HSD to the 3.11 fit with

`TukeyHSD(fit311, "mixing", conf.level = 0.95)` gives

Comparison	diff	lwr	upr	p_{adj}
2 - 1	185.25	-52.50	423.00	0.1494
3 - 1	-37.25	-275.00	200.50	0.9653
4 - 1	-304.75	-542.50	-67.00	0.0116
3 - 2	-222.50	-460.25	15.25	0.0693
4 - 2	-490.00	-727.75	-252.25	0.0003
4 - 3	-267.50	-505.25	-29.75	0.0262

At $\alpha = 0.05$, the only significant differences are those involving technique 4 (4 vs 1, 4 vs 2, 4 vs 3). Techniques 1, 2, and 3 are not significantly different from each other. This matches the idea from the mean-CI plot that 4 is clearly worse; Fisher's LSD in 3.11 flagged more pairs because it is less conservative.



(b) Fisher's LSD uses the ANOVA MSE and then runs ordinary pairwise t -tests at level α , without adjusting for the number of comparisons, so the familywise type I error can be much larger than α when there are many pairs. Tukey's HSD uses the studentized range distribution to give simultaneous confidence intervals for all pairwise differences that control the familywise error at α , so it is more conservative and usually declares fewer differences significant.

3. 3.14 A product developer is investigating the tensile strength of a new synthetic fiber that will be used to make cloth for men's shirts. Strength is usually affected by the percentage of cotton used in the blend of materials for the fiber. The engineer conducts a completely randomized experiment with five levels of cotton content and replicates the experiment five times. The data are shown in the following table:

Cotton weight percent	Observations				
15	7	7	15	11	9
20	12	17	12	18	18
25	14	19	19	18	18
30	19	25	22	19	23
35	7	10	11	15	11

- Is there evidence to support the claim that cotton content affects the mean tensile strength? Use $\alpha = 0.05$.
- Use the Fisher LSD method to make comparisons between the pairs of means. What conclusions can you draw?
- Analyze the residuals from this experiment and comment on model adequacy.

Solution: We fit a one-way ANOVA with factor “cotton percent”:

$$Y_{ij} = \mu + \tau_i + \varepsilon_{ij}, \quad i \in \{15, 20, 25, 30, 35\}, \quad j = 1, \dots, 5.$$

(a) Hypotheses:

$$H_0 : \mu_{15} = \mu_{20} = \mu_{25} = \mu_{30} = \mu_{35} \quad \text{vs} \quad H_a : \text{at least one mean differs.}$$

From `aov(strength ~ cotton)` we get

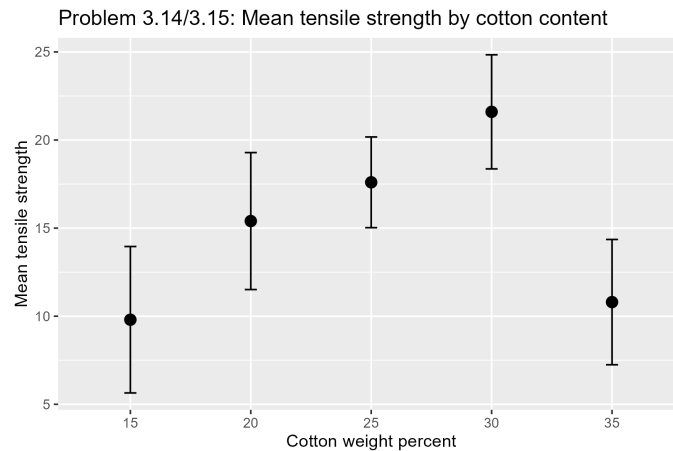
Source	Df	SS	MS	F
Cotton	4	475.8	118.94	14.76
Error	20	161.2	8.06	

with $p = 9.13 \times 10^{-6} < 0.05$. So cotton content affects mean tensile strength.

(b) The sample means are

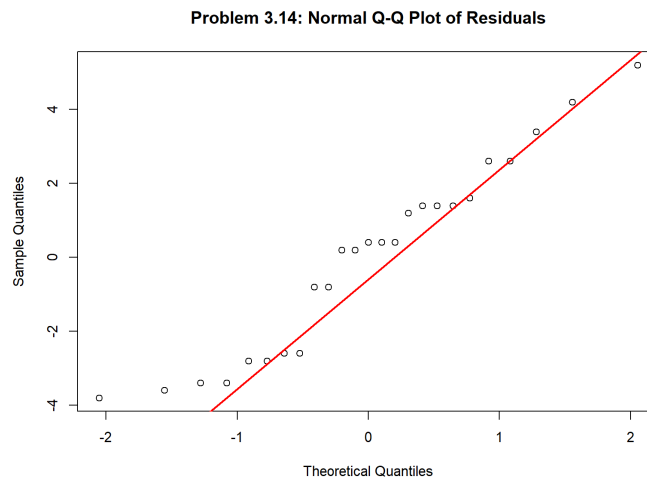
$$\bar{y}_{15} = 9.8, \quad \bar{y}_{20} = 15.4, \quad \bar{y}_{25} = 17.6, \quad \bar{y}_{30} = 21.6, \quad \bar{y}_{35} = 10.8.$$

Using $MSE = 8.06$ and $df_E = 20$, Fisher’s LSD pairwise tests (pooled t -tests, no adjustment) together with the mean–CI plot show the following pattern.



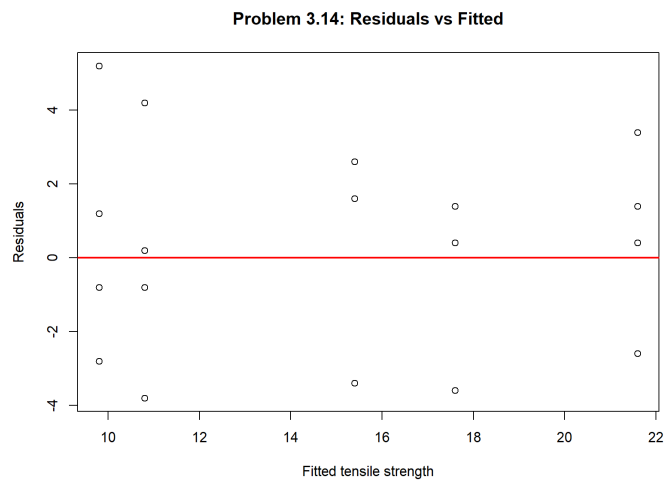
At $\alpha = 0.05$, 30% is significantly higher than 15%, 20%, and 35%; 20% is higher than 15% and 35%; 20% and 25% do not differ significantly, and 15% and 35% also do not differ. So 30% is best, 20–25% are intermediate, and 15% and 35% are clearly lower.

(c) The normal Q–Q plot of residuals is shown below.



The points stay close to the line with only mild tail deviations, so normality looks reasonable.

The residuals vs fitted plot is:



Residuals are roughly symmetric around zero with no clear trend or change in spread, so constant variance and the one-factor model both seem adequate.

4. 3.15 Reconsider the experiment described in Problem 3.14. Suppose that 30 percent cotton content is a control. Use Dunnett's test with $\alpha = 0.05$ to compare all of the other means with the control.

Solution: We reuse the one-way ANOVA fit from 3.14 and treat 30% cotton as the control. Let μ_{30} be the mean at 30%, and μ_i the mean at $i \in \{15, 20, 25, 35\}$. Dunnett's procedure tests

$$H_{0i} : \mu_i = \mu_{30} \quad \text{vs} \quad H_{ai} : \mu_i \neq \mu_{30}$$

for each noncontrol level, while controlling the familywise error at $\alpha = 0.05$.

From 3.14,

$$MSE = 8.06, \quad df_E = 20,$$

and the sample means are

$$\bar{y}_{15} = 9.8, \quad \bar{y}_{20} = 15.4, \quad \bar{y}_{25} = 17.6, \quad \bar{y}_{30} = 21.6, \quad \bar{y}_{35} = 10.8,$$

with $n_i = n_c = 5$ at all levels. The estimated differences and standard errors for $(\mu_i - \mu_{30})$ are

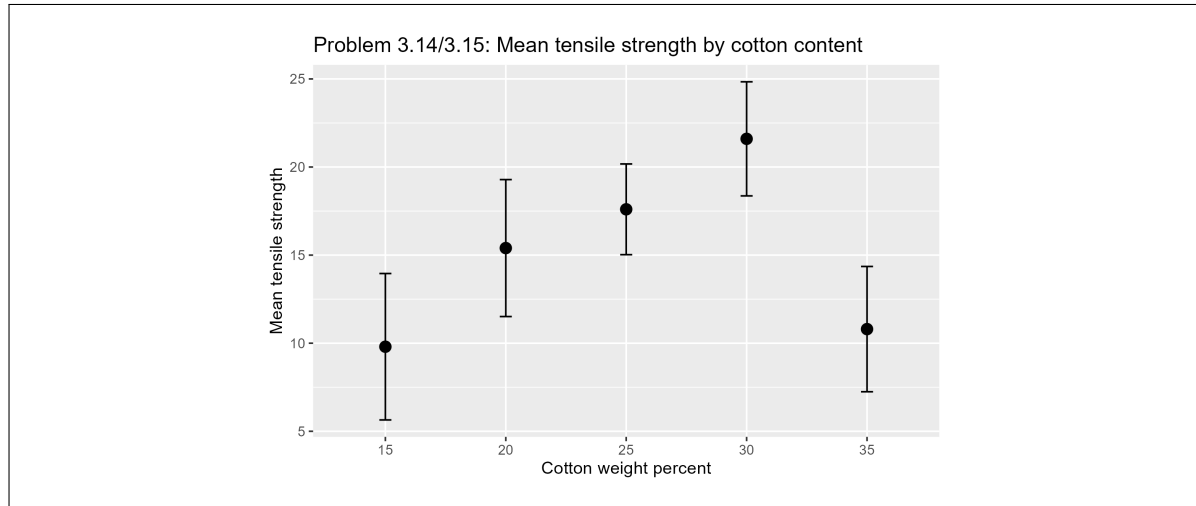
$$\hat{\mu}_i - \hat{\mu}_{30} = \bar{y}_i - \bar{y}_{30}, \quad SE(\hat{\mu}_i - \hat{\mu}_{30}) = \sqrt{MSE \left(\frac{1}{n_i} + \frac{1}{n_c} \right)} \approx 1.80.$$

The resulting table (from R) is

Cotton %	$\hat{\mu}_i - \hat{\mu}_{30}$	SE	t	Lower	Upper	Sig?
15	-11.8	1.80	-6.57	-16.71	-6.89	Yes
20	-6.2	1.80	-3.45	-11.11	-1.29	Yes
25	-4.0	1.80	-2.23	-8.91	0.91	No
35	-10.8	1.80	-6.01	-15.71	-5.89	Yes

Here the “Lower” and “Upper” columns are simultaneous 95% Dunnett-style confidence limits based on a simulated Dunnett critical value $\hat{c}_D \approx 2.73$ for $a = 5$ (one control, four treatments) and $df_E = 20$. A comparison is significant at $\alpha = 0.05$ if its interval excludes zero.

Thus 15%, 20%, and 35% all have significantly lower mean strength than the 30% control. The 25% level is somewhat lower but not significantly different from 30% once the multiple comparisons to the control are accounted for. This supports 30% as the best choice, with 25% as the only nearby competitor.



5. 3.22 A manufacturer of television sets is interested in the effect on tube conductivity of four different types of coating for color picture tubes. A completely randomized experiment is conducted and the following conductivity data are obtained:

Coating type	Conductivity			
1	143	141	150	146
2	152	149	137	143
3	134	136	132	127
4	129	127	132	129

- Is there a difference in conductivity due to coating type? Use $\alpha = 0.05$.
- Estimate the overall mean and the treatment effects.
- Compute a 95 percent confidence interval estimate of the mean of coating type 4. Compute a 99 percent confidence interval estimate of the mean difference between coating types 1 and 4.
- Test all pairs of means using the Fisher LSD method with $\alpha = 0.05$.
- Use the graphical method discussed in Section 3.5.3 to compare the means. Which coating type produces the highest conductivity?
- Assuming that coating type 4 is currently in use, what are your recommendations to the manufacturer? We wish to minimize conductivity.

Solution: We fit a one-way fixed-effects ANOVA with factor “coating type”:

$$Y_{ij} = \mu + \tau_i + \varepsilon_{ij}, \quad i = 1, \dots, 4, \quad j = 1, \dots, 4,$$

with independent $N(0, \sigma^2)$ errors.

(a) Hypotheses:

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4 \quad \text{vs} \quad H_a : \text{not all } \mu_i \text{ are equal.}$$

From `aov(conductivity ~ coating)` we obtain

Source	Df	SS	MS	F
Coating	3	844.7	281.56	14.3
Error	12	236.3	19.69	

with $p = 0.000288 < 0.05$. So coating type affects mean conductivity.

(b) The overall mean is

$$\bar{y}_{..} \approx 137.94.$$

The sample means are

$$\bar{y}_1 = 145.00, \quad \bar{y}_2 = 145.25, \quad \bar{y}_3 = 132.25, \quad \bar{y}_4 = 129.25,$$

so the estimated effects $\hat{\tau}_i = \bar{y}_i - \bar{y}_{..}$ are

$$\hat{\tau}_1 = 7.0625, \quad \hat{\tau}_2 = 7.3125, \quad \hat{\tau}_3 = -5.6875, \quad \hat{\tau}_4 = -8.6875.$$

Coatings 1–2 raise conductivity relative to the overall mean, 3–4 lower it.

(c) From the ANOVA, $MSE = 19.6875$ and $df_E = 12$, with $n_i = 4$ for each coating. For coating 4, $\bar{y}_4 = 129.25$, so a 95% CI for μ_4 is

$$\bar{y}_4 \pm t_{0.975,12} \sqrt{\frac{MSE}{n_4}} = 129.25 \pm 2.179 \sqrt{\frac{19.6875}{4}} = (124.42, 134.08).$$

For the mean difference $\mu_1 - \mu_4$, the estimate is $\bar{y}_1 - \bar{y}_4 = 15.75$ with

$$SE(\bar{y}_1 - \bar{y}_4) = \sqrt{MSE \left(\frac{1}{n_1} + \frac{1}{n_4} \right)}.$$

Using $t_{0.995,12} \approx 3.055$ for a 99% CI,

$$(\bar{y}_1 - \bar{y}_4) \pm t_{0.995,12} SE(\bar{y}_1 - \bar{y}_4) = (6.17, 25.33).$$

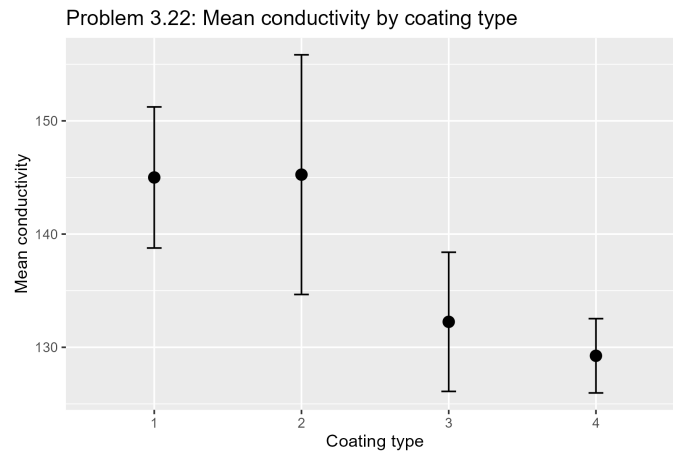
So coating 1 has clearly higher mean conductivity than coating 4.

(d) Fisher's LSD tests (pooled t -tests, no adjustment) based on MSE give the following p -values:

	1	2	3
2	0.93780		
3	0.00157	0.00136	
4	0.00030	0.00026	0.35785

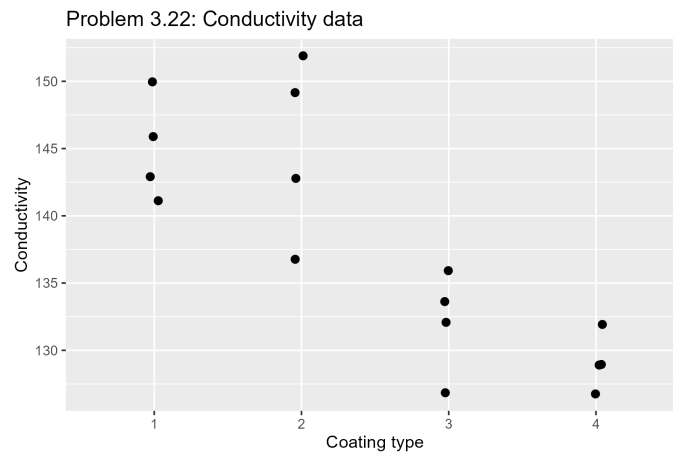
At $\alpha = 0.05$, coatings 1 vs 3, 1 vs 4, 2 vs 3, and 2 vs 4 are significantly different; 1 vs 2 and 3 vs 4 are not. Coatings 1–2 form a high-conductivity group, coatings 3–4 a low-conductivity group.

- (e) The mean–CI plot below shows the same pattern: 1 and 2 have the largest means, with intervals well above those of 3 and 4, and 3–4 are similar.



Coating 2 has the highest sample mean, very slightly above 1.

- (f) The scatter plot and the ANOVA both show that coatings 3 and 4 give lower conductivity than 1 and 2.



Since we want to minimize conductivity and coating 4 is currently used, note that 3 and 4 form the low group and are not significantly different. There is no strong statistical reason to prefer 3 over 4, but if a small additional reduction is considered worthwhile and switching is cheap, coating 3 is a reasonable candidate; otherwise staying with 4 is also defensible.

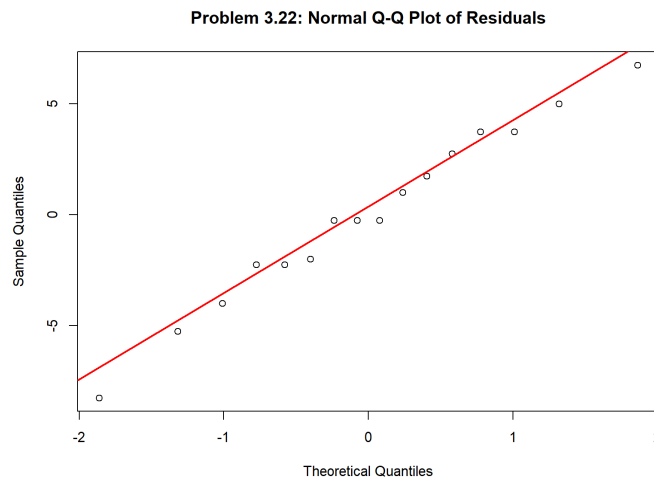
6. 3.23 Reconsider the experiment from Problem 3.22. Analyze the residuals and draw conclusions about model adequacy.

Solution: We use the one-way ANOVA model from 3.22 and examine residuals

$$\hat{\varepsilon}_{ij} = Y_{ij} - \hat{Y}_{ij}$$

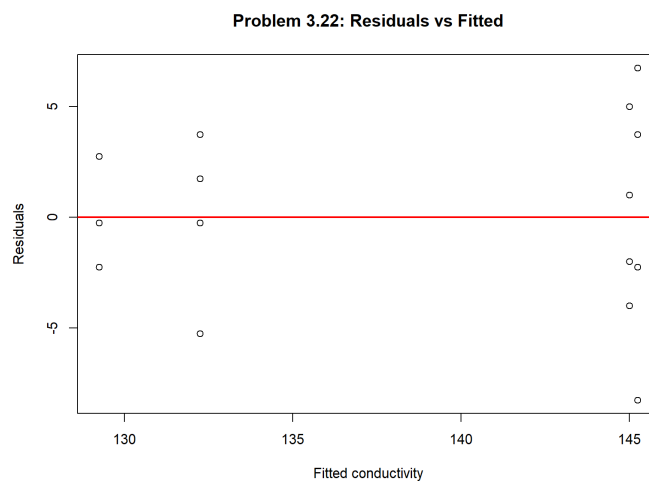
to check normality, constant variance, and model form.

The normal Q-Q plot of residuals is:



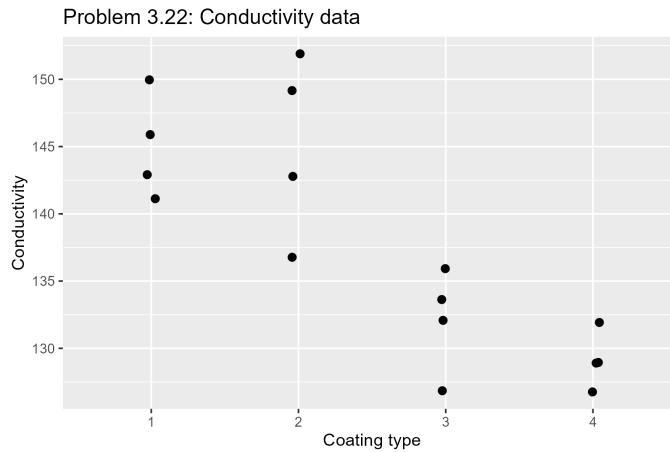
The points track the reference line closely, with only mild tail deviations, so the normality assumption looks fine.

Residuals vs fitted values:



Residuals are scattered around zero with no visible trend or change in spread, suggesting constant variance and no obvious lack of fit.

The raw-data plot by coating type:



shows no extreme outliers within any group and a clear separation between the high-conductivity (1–2) and low-conductivity (3–4) coatings. Overall, the residual analysis does not reveal problems with the one-factor ANOVA model used in 3.22.

7. 3.26 The response time in milliseconds was determined for three different types of circuits that could be used in an automatic valve shutoff mechanism. The results from a completely randomized experiment are shown in the following table:

Circuit type	Response time (ms)				
1	9	12	10	8	15
2	20	21	23	17	30
3	6	5	8	16	7

- (a) Test the hypothesis that the three circuit types have the same response time. Use $\alpha = 0.01$.
- (b) Use Tukey's test to compare pairs of treatment means. Use $\alpha = 0.01$.

Solution: We fit a one-way ANOVA with factor “circuit type”:

$$Y_{ij} = \mu + \tau_i + \varepsilon_{ij}, \quad i = 1, 2, 3, \quad j = 1, \dots, 5.$$

- (a) Hypotheses:

$$H_0 : \mu_1 = \mu_2 = \mu_3 \quad \text{vs} \quad H_a : \text{not all } \mu_i \text{ are equal.}$$

From `aov(time ~ circuit)` we get

Source	Df	SS	MS	F
Circuit	2	543.6	271.8	16.08
Error	12	202.8	16.9	

with $p = 0.000402 < 0.01$. So circuit type affects mean response time at the 1% level.

- (b) Tukey's HSD at familywise $\alpha = 0.01$ (`TukeyHSD(fit326, "circuit", conf.level=0.99)`) gives

Comparison	diff	lwr	upr	p_{adj}
2 - 1	11.4	2.12	20.68	0.00237
3 - 1	-2.4	-11.68	6.88	0.63670
3 - 2	-13.8	-23.08	-4.52	0.00050

At $\alpha = 0.01$, circuits 1 and 2 differ, and 2 and 3 differ; 1 and 3 do not. Circuit 2 has much larger mean response time than 1 and 3, which are similar, so if we want to minimize response time we would choose circuit 1 or 3 rather than 2.

8. 3.36 An article in the *Journal of the Electrochemical Society* (Vol. 139, No. 2, 1992, pp. 524–532) describes an experiment to investigate the low-pressure vapor deposition of polysilicon. The experiment was carried out in a large-capacity reactor at Sematech in Austin, Texas. The reactor has several wafer positions, and four of these positions are selected at random. The response variable is film thickness uniformity. Three replicates of the experiment were run, and the data are as follows:

Wafer position	Uniformity		
1	2.76	5.67	4.49
2	1.43	1.70	2.19
3	2.34	1.97	1.47
4	0.94	1.36	1.65

- (a) Is there a difference in the wafer positions? Use $\alpha = 0.05$.
 (b) Estimate the variability due to wafer positions.
 (c) Estimate the random error component.
 (d) Analyze the residuals from this experiment and comment on model adequacy.

Solution: We use a one-way random-effects ANOVA:

$$Y_{ij} = \mu + \alpha_i + \varepsilon_{ij}, \quad i = 1, \dots, 4, \quad j = 1, 2, 3,$$

where $\alpha_i \sim N(0, \sigma_\alpha^2)$ is the random wafer effect and $\varepsilon_{ij} \sim N(0, \sigma^2)$, independent.

- (a) To test for position-to-position variability we test

$$H_0 : \sigma_\alpha^2 = 0 \quad \text{vs} \quad H_a : \sigma_\alpha^2 > 0.$$

From `aov(uniformity ~ wafer)` we obtain

Source	Df	SS	MS	<i>F</i>
Wafer	3	16.220	5.4066	8.29
Error	8	5.217	0.6522	

with $p = 0.00775 < 0.05$. So wafer position does contribute variability in uniformity.

(b) In this model,

$$E(MS_{\text{wafer}}) = \sigma^2 + r\sigma_{\alpha}^2, \quad E(MS_E) = \sigma^2, \quad r = 3.$$

So

$$\hat{\sigma}_{\alpha}^2 = \frac{MS_{\text{wafer}} - MS_E}{r} = \frac{5.406608 - 0.6521833}{3} \approx 1.58.$$

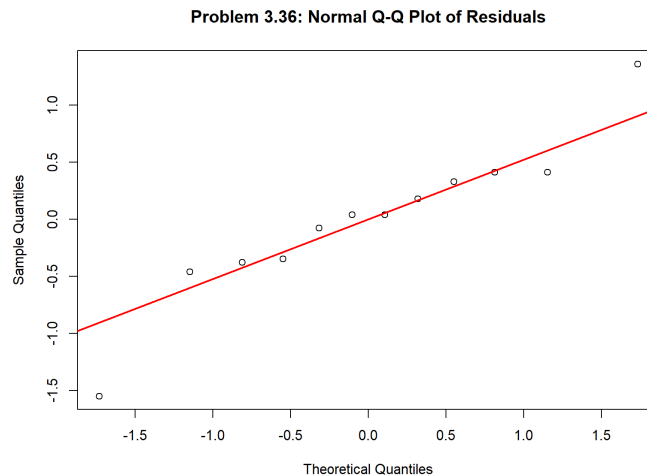
This is the estimated variance due to wafer-to-wafer differences.

(c) The error variance component is estimated by

$$\hat{\sigma}^2 = MS_E = 0.6521833,$$

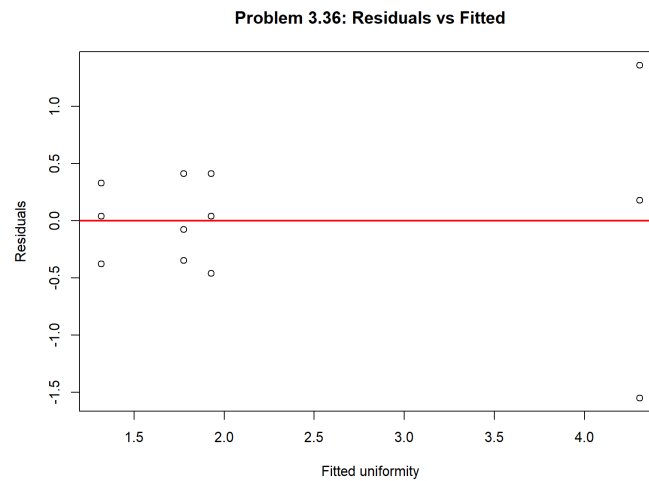
so the within-position variance is about 0.65.

(d) The normal Q-Q plot of residuals is



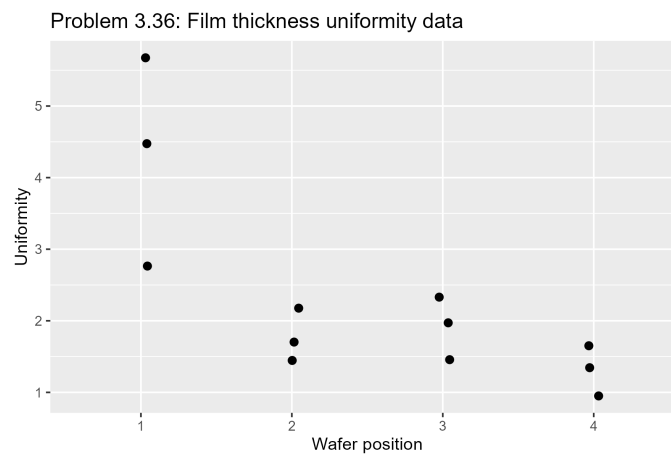
which is reasonably straight with only small tail deviations, so normality looks acceptable.

Residuals vs fitted values:



show residuals scattered around zero with roughly constant spread, suggesting the equal-variance assumption is fine.

The raw-data plot by position



shows positions 1 and 3 tending higher and 2 and 4 lower, matching the ANOVA. Overall, the one-way random-effects model seems adequate.

9. 3.52 Suppose that four normal populations have means of $\mu_1 = 50$, $\mu_2 = 60$, $\mu_3 = 50$, and $\mu_4 = 60$. How many observations should be taken from each population so that the probability of rejecting the null hypothesis of equal population means is at least 0.90? Assume that $\alpha = 0.05$ and that a reasonable estimate of the error variance is $\sigma^2 = 25$.

Solution: We have $a = 4$ normal populations with means

$$\mu_1 = 50, \mu_2 = 60, \mu_3 = 50, \mu_4 = 60,$$

common error variance $\sigma^2 = 25$, and ANOVA level $\alpha = 0.05$. Let n be the common sample size per group.

For a one-way ANOVA, the noncentrality parameter under a given alternative is

$$\lambda = \frac{n}{\sigma^2} \sum_{i=1}^a (\mu_i - \mu)^2,$$

where μ is the grand mean. Here

$$\mu = \frac{50 + 60 + 50 + 60}{4} = 55, \quad \sum_{i=1}^4 (\mu_i - \mu)^2 = 100,$$

so

$$\lambda = \frac{n \cdot 100}{25} = 4n.$$

The F -test has $df_1 = a - 1 = 3$ and $df_2 = a(n - 1) = 4(n - 1)$, with critical value

$$F_{\text{crit}} = F_{3, 4(n-1), 1-\alpha}.$$

For each n we can compute the power

$$\text{Power}(n) = P(F_{3, 4(n-1)}(\lambda) > F_{\text{crit}}),$$

where the F has noncentrality $\lambda = 4n$. In R:

```
mu <- c(50, 60, 50, 60)
sigma2 <- 25
alpha <- 0.05
a <- length(mu)

compute_power <- function(n) {
  grand_mean <- mean(mu)
  lambda <- n * sum((mu - grand_mean)^2) / sigma2
  df1 <- a - 1
  df2 <- a * (n - 1)
  F_crit <- qf(1 - alpha, df1, df2)
  1 - pf(F_crit, df1, df2, ncp = lambda)
}

n_values <- 2:50
powers <- sapply(n_values, compute_power)
```

The relevant part of the table is

n	Power(n)
2	0.2987
3	0.6157
4	0.8224
5	0.9270
6	0.9725

The smallest n with power at least 0.90 is $n = 5$ (power ≈ 0.927). Thus we need 5 observations per population.